

Understanding the Impact of Technical Issues on People's Perception and Attribution of Responsibility in Videoconferencing

Chi-Lan Yang

Graduate School of Interdisciplinary Information Studies,
The University of Tokyo
Tokyo, Japan
chilan.yang@cyber.t.u-tokyo.ac.jp

Takuji Narumi

Graduate School of Engineering, The University of Tokyo
Tokyo, Japan
narumi@cyber.t.u-tokyo.ac.jp

Xiaotong Li

Graduate School of Interdisciplinary Information Studies,
The University of Tokyo
Tokyo, Japan
xiaotong@cyber.t.u-tokyo.ac.jp

Hideaki Kuzuoka

Graduate School of Information Science and Technology,
The University of Tokyo
Tokyo, Japan
kuzuoka@cyber.t.u-tokyo.ac.jp

ABSTRACT

When interacting with people online, unexpected technical issues such as network impairments or background noise happen frequently. Though studies have shown that network outages negatively impacted remote collaboration, we know little about (1) how people attribute the cause of these technical issues and (2) how technical issues affect people's perception of their remote interlocutors and the communication system. In an online controlled experiment with 118 participants, we compared people's perception of the remote speaker and the communication system with and without experiencing technical issues in the simulated online interaction. The result showed that people perceived the remote speaker as less suitable and less credible when experiencing technical issues than without experiencing technical issues in videoconferencing. Moreover, people were not aware that their assessment of the speaker was affected by technical issues. We highlight the need to enhance people's awareness about the impact of uncontrollable technical issues in remote communication.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**.

KEYWORDS

Computer-mediated communication, attribution, perception, videoconferencing, technical issues

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1 INTRODUCTION

Technical issues such as network outages, connection loss caused by video and audio misoperation, and unexpected background noise were prevalent when more and more social interactions moved online. The inevitable technical issues in remote communication have been found to have negative impacts on user experiences [8] and collaboration outcome [1]. However, we know little about how people's perception of their remote counterparts would be affected by unexpected and often uncontrollable technical issues. In online social interactions, people could only rely on limited cues transmitted over computer-mediated communication (CMC) technology to form impressions of each other [18]. When experiencing technical issues during serious occasions such as online job interview or online court, where first impression is critical for the whole social interactions, it is essential to know how people attribute the cause of technical issues and how their perception of remote speakers will be affected in the middle of video-mediated communication.

Humans tend to assign a cause to an observed outcome, infer traits by observing behaviors, or ascribe blame to a person [13]. This attribution process occurs when people try to build trust and understand each other based on the information they collected during the social interactions [9, 10]. In CMC, remote interlocutors could only rely on limited social cues to infer others' behavior when building trust online. As a consequence, people are likely to associate the observed behavior with a wrong cause [7]. Such misattribution happens when people explain the cause based on limited information, leading people to infer the cause of the observed outcome incorrectly to something unrelated [3, 6].

Previous studies have shown that when attributing the cause of unfavorable outcomes between human and intelligent systems, people tend to think that humans should take more responsibility for the failure than the computer and intelligent systems [5, 14]. Yet,

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we know little about how people attribute the unfavorable experiences when having social interactions over CMC. When interacting with people in videoconferencing, various possible reasons can lead to unexpected technical issues, and it is hard to identify the exact cause of the problem. For example, when experiencing an unstable internet connection, it can be caused by one's own internet provider, one's own computer, the server of the communication system, remote counterpart's internet provider, or any other unknown possible factors. When such unexpected technical issues happen, how would people attribute the responsibility of the unfavorable experience?

As carrying out job interviews through video conferencing is prevalent, it is essential to know how technical issues affect peoples' perception of the remote counterpart during online interactions. This exploratory study compared three types of technical issues with a baseline (no technical issues), including personal, environmental, and multiple technical issues during people's first online encounter through videoconferencing. We also tested the impact of technical issues with three different workplace speech contents: professional-formal speech, non-professional-formal speech, and casual speech. We chose these three speech contents because both formal and casual conversations are frequently held through videoconferencing nowadays. We focused on the recipient's perception of the remote speaker and attribution of responsibility when experiencing technical issues during videoconferencing. Specifically, we ask three research questions:

- RQ1: When experiencing technical issues in formal and casual videoconferencing, how do technical issues influence people's perception of their remote speaker, in terms of suitability and credibility?
- RQ2: When experiencing technical issues in formal and casual videoconferencing, how do technical issues influence people's perception of the communication system, in terms of credibility?
- RQ3: When experiencing technical issues in formal and casual videoconferencing, how do people attribute the responsibility of technical issues?

2 BACKGROUND AND MOTIVATION

When communicating via computer-mediated communication (CMC) technologies, people usually suffer from technical issues like network impairment and unexpected noise. Ahsen *et al.* reported that such technical issues in remote communication could cause frustration [1]. In many cases, such technical issues are caused by limitations of the communication infrastructure, such as low bandwidth of the network and network delay. While in some other cases, communication troubles are caused due to each person's local problems, such as environmental noise or the bad connection of a headset.

When encountering the above-mentioned technical issues during videoconferencing, it is unclear how people attribute the cause of the poor communication experience. According to attribution theory, human tends to explain the outcomes they observed with specific reasons, even when the real causes of the result were uncertain [11]. This attribution process can be biased by humans' tendency to underestimate the influence of situational factors and

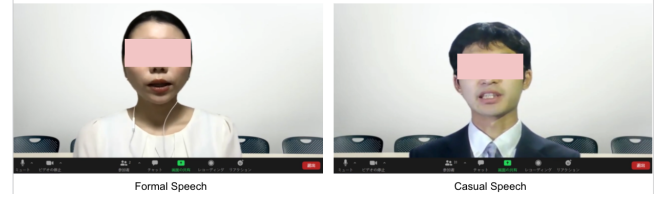


Figure 1: The left screenshot illustrates a female speaker performing a formal speech, and the right screenshot shows a male speaker performing a casual speech. Both female and male speakers recorded professional, non-professional, and casual speech.

jump too quickly to the conclusion of dispositional factors [6]. For example, when seeing someone being jobless, people tend to assume the person is lazy or lacks competency without considering the influence of various social contexts. Several factors could lead to *misattribution*, which indicates that people incorrectly link the outcomes to unrelated causes [4, 6].

When having social interactions over CMC, the limited cues people received from communication technologies could lead people to form incorrect assumptions or impressions of remote counterparts. For example, misinterpreting the smile as being polite during an online discussion [7], or associating the act of turning the video off with low engagement in videoconferencing [12]. Insufficient available nonverbal information transmitted over remote communication makes people overemphasize certain characteristics based on the limited cues they received [19]. When experiencing technical issues during the first online encounter, would people form an incorrect impression of the remote speaker due to technical issues?

There are several reasons to cause technical issues. One can attribute it to the artifact (e.g., network environment, communication device) or the person who interacted with the artifact. Related works have shown that when experiencing failure in a collaborative game mediated by a computer, people tend to attribute the cause of failure to the other human partner, rather than the computer or robot partner [14]. Elish *et al.* also revealed that people tend to blame the human operator for the failure operation even if the action was largely automated by computers [5]. Therefore, we wonder when experiencing technical issues in computer-mediated communication, how people perceive both speakers (RQ1) and the communication system (RQ2), and how people attribute the responsibility of technical issues (RQ3).

3 METHOD

3.1 Study Design

To investigate the influence of various technical issues on people's perception, we compared three types of technical issues with a baseline condition ($T_{baseline}$ without technical issues), including personal ($T_{personal}$), environmental ($T_{environmental}$), and multiple ($T_{multiple}$) technical issues. In total, we got four groups. We identified the above three technical issues because they frequently occur in online meetings. We defined audio disconnection as a controllable personal issue. In $T_{personal}$, we controlled the volume to make the sound loud, low and muted in different parts of the video.

While people could not control the unstable internet connection or background noise, we defined them as environmental technical issues. In $T_{environmental}$, we simulated the frozen and choppy effect as the unstable connection so that the video and sound appeared unsynchronized. We also added the noise from the construction site in the background. These effects were embedded in different parts of the video. For $T_{multiple}$ condition, we added both personal and environmental technical issues to the video. We inserted three types of simulated technical issues into the same part of the speech. We inserted them to the part where the original meaning of the speech can still be inferred correctly from the context even without these utterances. For example, [this] is John; Welcome [to] join the event. The percentage of unheard or unclear utterances among each speech was less than 38%. (The example clip can be seen in the supplementary materials.)

We were also curious about the influence of technical issues under different speech contents. Therefore, we designed three prevalent workplace communication scenarios [2], including a professional interview, non-professional interview, and hosting a casual conversation. In both professional and non-professional speech, we invited a male and a female confederate to play as the interviewee for making a 30-second short self-introduction in an online job interview. We added a Zoom-like menu bar at the bottom of the video. We set the number of meeting participants to two to simulate the online job interview, where the participant was the interviewer, and the speaker in the video was the interviewee. The speaker of the professional speech showed enthusiasm and appeared competitive in the self-introduction. The speaker of the non-professional seemed less motivated and not competitive in getting the job. For the casual setting, the same male and female confederates were instructed to introduce an agenda for an online social event. We added a menu bar on the bottom of the ZOOM, showing that 31 attendees were in this virtual room to simulate it with the actual online social event. We provided the confederates with the same transcripts for recording the three speech contents and instructed them to use a neutral tone and facial expressions. They wore formal suits and set the same virtual background assigned by the researchers while recording (Fig. 1).

All participants were randomly assigned to one of the four technical issue groups, and all experienced three types of communication topics. For example, if participants were assigned to $T_{personal}$ condition, they would watch professional, non-professional, and casual speech contents, and all videos were introduced with personal technical issues. Each participant watched all three pre-recorded 30-second videos in random order.

3.2 Participants

We recruited 141 participants (56 females, 85 males) with the mean age of 41.5 years old ($SD = 9.1$) from a local Japanese participant recruiting platform¹. All participants were paid \$1 USD for their participation. We decided the amount based on the standard hourly wage in the country where the study was conducted. We recruited participants by telling them this was an online survey that required them to assess someone in the assigned short video clip. The ethical review board of the authors' institute approved the study.

¹We recruited participants from Lancers.jp, an online freelancing platform.

3.3 Procedure

The study was carried out with an online survey with different videos embedded in the survey. Participants were randomly assigned to four different conditions, $T_{baseline}$, $T_{personal}$, $T_{environmental}$, and $T_{multiple}$. To make participants experience three speech contents, we guide them with a formal and casual workplace scenario. For professional and non-professional speech, we told the participants that they were the interviewers of a company, and their task was to judge whether the remote speaker was suitable for the job position. For casual speech, participants had to evaluate whether the speaker in the video organized the social event well. Each participant watched videos with three different types of speech content: professional, non-professional, and casual. Participants only experienced one technical issue for the three speech contents depending on their assigned conditions. The gender of the speaker in the video was counterbalanced.

In the survey, first, we collected participants' demographic information. Next, participants watched a video and then evaluated the person and remote communication system in the video. In the evaluation phase, participants assessed the remote speaker's suitability (*How much do you think the speaker is suitable for this job?*), the credibility of the speaker, and the communication system (*I think the speaker/communication system is: Intelligent, Competent, Honest, Expert, Bright, Trustworthy*) [16]. Participants answered all survey items with a 7-point Likert scale, ranging from 1 (totally disagree) to 7 (totally agree).

Afterward, participants who were assigned to conditions with technical issues were asked whether they noticed any technical issues while watching the video, and if yes, how they thought about the cause of the observed issues. The questions were: *When tech issues happened, I thought the speaker should be responsible for it*; *When tech issues happened, I thought the communication system should be responsible for it*; *When tech issues happened, I thought neither the speaker nor the communication system should be responsible for it*². We designed all items with a 7-point Likert scale, where one indicated strongly disagree, and seven indicated strongly agree. We treated this score as *attribution score* in the analysis.

4 RESULTS

To validate whether participants paid attention to the video clip, we asked them to answer a validation question about the content of the video. As a result, we removed 23 participants who did not pass the validation question and those who partially completed the study. In total, our analysis was based on the data from 118 participants ($T_{baseline}$: $n=28$, $T_{personal}$: $n=31$, $T_{environmental}$: $n=29$, $T_{multiple}$: $n=30$).

4.1 The Effect of Technical Issues on the Assessment of the Remote Speaker (RQ1)

To answer RQ1, we performed a one-way ANOVA³ to investigate the effect of technical issues on the perceived job suitability and credibility of the remote speaker in videoconferencing. The result showed that when the remote speaker delivered a professional

²This is the reverse item.

³We did not use a two-way ANOVA because the interaction effect between types of technical issues and types of speech was not our research interests.

speech while having multiple technical issues, participants gave the lowest suitability score than baseline condition ($F[3, 114]=4$, $p<.01$, $Cohen's f=0.36$). Result of post-hoc analysis using LSD multiple comparison revealed that when experiencing environmental ($T_{environmental}$: $M=4.76$, $SD=1.10$) and multiple technical issues ($T_{multiple}$: $M=4.6$, $SD=0.92$), participants rated the remote speakers as less suitable for the job than without having any technical issues ($T_{baseline}$: $M=5.61$, $SD=0.98$). However, when remote speakers were delivering a non-professional and casual speech contents, we did not find significant effects of three types of technical issues on participants' perceived suitability toward the speaker, compared with baseline condition (Non-professional: $F[3, 114]=2.28$, $n.s.$, $Cohen's f=0.25$; Casual: $F[3, 114]=1.12$, $n.s.$, $Cohen's f=0.17$) (Fig. 2).

Regarding participants' assessment of remote speakers' credibility, we averaged six items of credibility measure and got a credibility score. Result of a one-way ANOVA showed that participants gave the lowest credibility rating to the remote speaker when experiencing multiple technical issues than baseline condition, when viewing both professional and non-professional speech content (Professional: $F[3, 114]=2.80$, $p<.05$, $Cohen's f=0.27$; Non-professional: $F[3, 114]=3.71$, $p<.05$, $Cohen's f=0.31$). Post-hoc analysis of LSD revealed that experiencing multiple technical issues ($T_{multiple}$ in professional speech: $M=5.16$, $SD=0.72$; $T_{multiple}$ in non-professional speech: $M=3.94$, $SD=0.53$) in videoconferencing made participants give lower credibility ratings to remote speakers than baseline condition ($T_{baseline}$ in professional speech: $M=5.65$, $SD=0.61$; $T_{baseline}$ in non-professional speech: $M=4.42$, $SD=0.77$). However, for casual speech, any types of technical issues did not affect the remote speakers' credibility rating ($F[3, 114]=2.59$, $n.s.$) (Fig. 3).

Taken together, when experiencing multiple technical issues in videoconferencing, people had a negative impression of remote counterparts' suitability and credibility, especially when the speech content is professional.

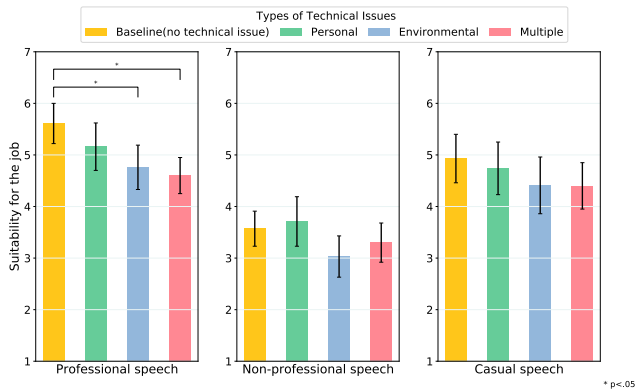


Figure 2: Mean ($\pm$ 95% confidence interval) of the suitability level (RQ1). X-axis shows three speech contents. Y-axis shows perceived job suitability. The higher, the more suitable. (* indicates $p < .05$)

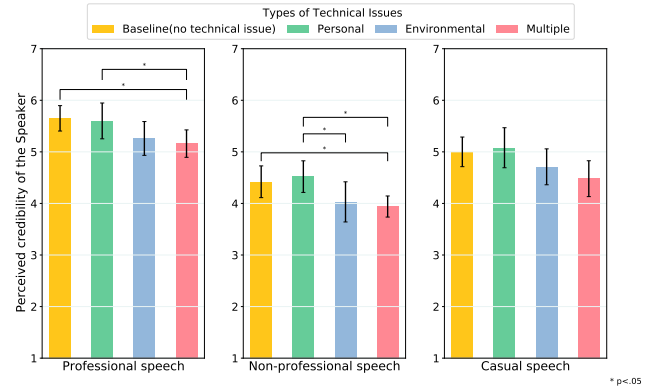


Figure 3: Mean ($\pm$ 95% confidence interval) of the perceived credibility. The X-axis shows three causes when experiencing multiple technical issues. Y-axis shows the perceived credibility of the speaker. A high score indicates participants perceived the speaker had high credibility. (* indicates $p < .05$)

4.2 The Effect of Technical Issues on the Assessment of the Communication System (RQ2)

Next, we turn to see how types of technical issues affect participants' perception of the communication system (RQ2). Result of a one-way ANOVA showed that types of technical issues had a negative impact on participants' perceived credibility of the communication system across three speech contents (Professional: $F[3, 114]=20.42$, $p<.01$, $Cohen's f=0.73$, Non-professional: $F[3, 114]=19.72$, $p<.01$, $Cohen's f=0.72$; Casual: $F[3, 114]=22.27$, $p<.01$, $Cohen's f=0.77$). Post-hoc analysis of LSD showed that no matter what types of technical issues were experienced, participants rated the credibility of communication system significantly lower than baseline condition (Fig. 4).

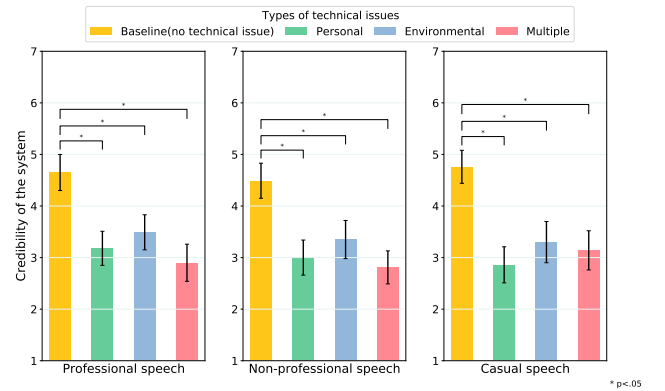


Figure 4: Mean ($\pm$ 95% confidence interval) of the perceived credibility of the communication system (RQ2). The X-axis shows three speech contents. Y-axis shows system credibility. The higher, the more credible. (* indicates $p < .05$)

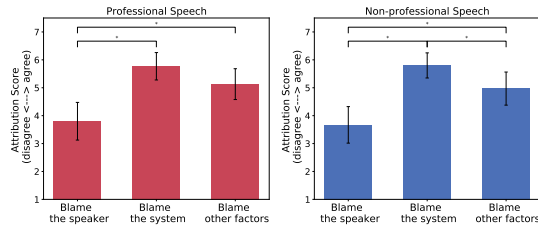


Figure 5: Mean ($\pm$ 95% confidence interval) of the attribution score (RQ3). The X-axis shows three causes when experiencing multiple technical issues. The Y-axis shows how much participants thought the source should be responsible for causing technical issues. A higher score indicates participants attribute more responsibility to the source. (* indicates $p < .05$) Left figure with red bar shows the video content was professional self-introduction, whereas right figure with blue bar shows the video content was non-professional self-introduction.

4.3 Attributing the Responsibility of Technical Issues (RQ3)

After knowing that multiple technical issues caused a negative impression of the remote speakers in both professional and non-professional speech contents (Sec. 4.1), we moved further to examine how participants explain the cause of technical issues (RQ3). We compared how much participants thought the speaker, the system, or other factors should be responsible for the technical issues. The result of a one-way ANOVA revealed that when experiencing multiple types of technical issues in professional and non-professional speech, there was a significant difference of how participants attributed to three different sources ($T_{multiple}$ in professional speech: $F[2, 58]=16.51, p<.01, \text{Cohen's } f=0.75$; $T_{multiple}$ in non-professional speech: $F[2, 58]=18.18, p<.01, \text{Cohen's } f=0.79$). Post-hoc analysis of LSD showed that when the video was professional self-introduction, participants thought the system should be responsible for the technical issues more than the speaker (system: $M=5.77, SD=1.33$; speaker: $M=3.8, SD=1.83$) (see Fig. 5, left). Similar trend was found when the video was about non-professional self-introduction. Post-hoc analysis of LSD showed that participants thought the system should be responsible for the technical issues more than the speaker (system: $M=5.8, SD=1.22$; speaker: $M=3.67, SD=1.76$) (see Fig. 5, right).

Even participants attributed the responsibility of technical issues to the system more than the speaker, their perceptions toward remote speakers were still negatively impacted. The above result suggested that participants were unaware that their perceived credibility of the speaker was affected by the technical issues they experienced.

5 DISCUSSION AND FUTURE DIRECTIONS

When encountering unexpected technical issues in videoconferencing, our results demonstrated that technical issues negatively impacted people's perception of the remote speakers and the communication systems. Moreover, when asked about the cause of the

problem, even though people thought the artifact should be responsible for the technical issue rather than the speaker (Sec. 4.3), their perception of the remote speaker was still negatively affected (Sec. 4.1).

This study revealed that technical issues had a significantly negative impact on speakers' credibility when conducting formal speeches (Fig. 3). However, this impact was not salient when the speaker performed casual speeches (Sec. 4.1). One possible explanation might be that the formal contexts led people to set strict criteria while evaluating. Thus, participants assumed the speaker should not make any mistake, even though people knew that technical issues were not the speakers' fault. However, there was no need to assume and judge one's competency when the situation was casual. Therefore, people weigh less about the video quality than the formal situation.

Interestingly, the current result showed that people did not become aware that their assessment of the speaker was affected by the technical issues. Prior studies have explored extensively on how people attribute of responsibilities during human-computer [15], human-robot [11], and human-AI interactions [10]. However, we had little understanding of how people attributed responsibilities when technologies mediated human social interactions. To reduce the negative impact of misattribution in computer-mediated communication (CMC), we need to consider both the computing system and people during attribution. According to related literature, increasing the transparency of the computing system could reduce the misattribution of the system [11]. On the other hand, increasing situation awareness could prevent people from overly relying on dispositional attributions when having negative stereotypes on others [17]. When encountering negative experiences over CMC, more studies are needed to explore how to redirect people's attribution of the responsibility away from the interlocutor and the computing system.

We acknowledge the study was limited in its generalisability of studying people from one cultural background. Despite this limitation, this exploratory study raised concerns when evaluating computer-mediated communication technologies. For critical events like job interviews or legal trials conducted online, it is essential to let people know that one's judgment can be influenced when experiencing unexpected technical issues over CMC. For our future work, we will focus on (1) increasing people's awareness of attribution bias and (2) designing feedback mechanisms to intervene in people's attribution process when experiencing technical issues in CMC to reduce the negative impact caused by technical issues.

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